Stepping Stone

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1. Research Question

Our research question for this assignment is "What is the relationship between perceived loneliness, social connection (number of close relationships), and digital media usage across global populations?". To answer this, our research will shed some light on the rise of the "loneliness epidemic" that is being noticed and being experienced by people around the world. In more detail, we will look into these datasets such as family value perceptions in various countries, how many friends or relatives people can count on. Combining with the statistics of internet population, daily hours spent with digital media. The research will be important for understanding how patterns of social connection are distributed and changing across populations. In the context of social data analytics, this research matters because it relies on quantifying subjective experiences like loneliness and comparing them to measurable social relationships and media usage. This can provide valuable insights for policymakers, mental health professionals, and platform designers seeking to improve on the social impacts of loneliness in the digital age.

2. Description of the Data

To answer this complex research question, we found several datasets to analyse. These datasets primarily come from articles published on *Our World in Data*, a website run by a nonprofit organization that aims to "advance education in how global living conditions … are changing" (Global Change Data Lab, n.d.). Our initial datasets come from the article *Loneliness and Social Connections* by Esteban Ortiz-Ospina (2020). The datasets that we are using from this article include; how important family is to people in life, and people who report having friends or relatives they can count on. Moreover, we also plan to use a dataset that illustrates the share of the population using the Internet, from the article *Internet* by Ritchie et al. (2023). These datasets

contain data that spans over several years and numerous countries, that are stored within comma-separated values files that are structured with the headers; country name, country code, year, and percentage of population. In addition to these datasets from *Our World in Data*, we also found a dataset from *Statisca* that we would like to analyze. This dataset, created by Dixon (2025), will help us perceive the daily amount of social media use across all countries during Q1 of 2023. It is stored within a portable document format file, and is structured as a horizontal bar graph, with the country's name being on the left, and the hours and minutes spent being on the right. In total, the four datasets that will be analyzed are around 241 kilobytes in size. By doing this brief analysis of the datasets that we will be using, we were able to identify clear usable data that may provide use to the final research project.

3. Justification of Data

These four datasets provide databases for analyzing the relationship between family values, social support, and digital behaviors. Through the "how important family is to people in life" dataset, we can understand the value placed on family in different national cultures. The second dataset, "People who report having friends or relatives they can count on" can directly measure the level of perceived social support in each country. The third dataset about the population using the Internet assists us in understanding the global penetration of digital technologies. The last dataset about average time people spend on social media per day reveals how active individuals are in the digital social space. Combining these data we can make cross-country comparisons and explore our research question. Also, analyze whether the Internet and social media use play a role in this.

There are several limitations. First, the time gap between these datasets. For example, social support data is from 2014–2016, family values data from 2022, and internet/social media

usage from 2023, which may affect temporal comparability. Second, different cultures will have different understandings of "family," "support," or "social media use," which may affect inconsistencies in responses. Third, not all countries appear in every dataset, we may lack some data. Finally, self-reported data may be influenced by social desirability bias, since people might overstate their responses.

4. Method(s) of Analysis

We plan to begin with correlation analysis to explore the relationship between loneliness, family value, social relationships and digital media usage. According to Emily James (n.d) on *Definition of Correlation Analysis* online blog, it is an approach to determine the strength and the "relationship between two variables/datasets" (para.2). This approach will help us identify whether variables such as the importance of family, having friends or relatives to rely on, and social media usage are positively or negatively associated across countries.

Additionally, we will conduct cross-national comparative analysis to examine how these patterns differ across various regions or cultural contexts. This approach is mentioned in Jacob Heath's et. al article (2024) involves studying a phenomenon in "two or more counties" in the goal of comparing the manifestation of the said phenomena (p.1).

Finally, we plan to use the Exploratory Data Analysis method to analyze our results. Data visualization methods such as scatter plots, bar charts are applied. For example, we can use scatter plots to clearly show the relationship between family importance and perceived social support.

These approaches align well with our research goal of understanding how the phenomenon of loneliness may emerge in different social contexts by extracting meaningful

patterns from the datasets and translating them into relevant insights to mental health and social behavior in a digital era.

References

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